

Team Name _____ Team # _____ Score _____

Student Names _____, _____

ALL ANSWERS MUST BE ON THE ANSWER SHEET (Answers on this test will not count).

Freshwater Aquatic Environments Test

Part I: Freshwater Systems, Ecology, and Human Impacts (Questions 1–40)

1. Which of the following best defines a freshwater aquatic environment?

- A. Water with high salt concentration
- B. Water with less than 1% dissolved salts
- C. Water found only underground
- D. Water that is frozen year-round

2. Lakes and ponds are classified as which type of freshwater system?

- A. Lotic
- B. Marine
- C. Lentic
- D. Estuarine

3. Rivers and streams are examples of which freshwater system?

- A. Lentic
- B. Lotic
- C. Benthic
- D. Pelagic

4. Which zone of a lake receives enough sunlight for photosynthesis?

- A. Profundal zone
- B. Benthic zone
- C. Aphotic zone
- D. Littoral zone

5. What is the primary source of energy for most freshwater ecosystems?

- A. Decomposers
- B. Heat from Earth's core
- C. Sunlight
- D. Chemical nutrients

6. Which process in the water cycle moves water from plants into the atmosphere?

- A. Condensation
- B. Evaporation
- C. Transpiration
- D. Precipitation

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7. Runoff is most likely to increase when land is:

- A. Covered with native vegetation
- B. Frozen
- C. Flat and grassy
- D. Paved with impervious surfaces

8. Which nutrient is most commonly responsible for freshwater eutrophication?

- A. Calcium
- B. Sodium
- C. Phosphorus
- D. Iron

9. Eutrophication often results in:

- A. Increased dissolved oxygen
- B. Algal blooms and oxygen depletion
- C. Clearer water
- D. Reduced plant growth

10. Which gas is essential for aquatic organisms that perform respiration?

- A. Carbon dioxide
- B. Nitrogen
- C. Oxygen
- D. Methane

11. Low dissolved oxygen levels in water are most likely to cause:

- A. Increased biodiversity
- B. Fish kills
- C. Higher pH
- D. Increased water clarity

12. The pH of most freshwater ecosystems typically ranges from:

- A. 1–3
- B. 4–6
- C. 6–8
- D. 9–11

13. Acid rain can lower the pH of freshwater systems, making it difficult for organisms to:

- A. Float
- B. Reproduce and survive
- C. Absorb sunlight
- D. Migrate

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14. Which chemical property of water helps moderate temperature changes in lakes?

- A. Density
- B. Polarity
- C. High specific heat
- D. Solubility

15. Which step is usually the FIRST in potable water treatment?

- A. Chlorination
- B. Filtration
- C. Coagulation
- D. Sedimentation

16. The primary purpose of chlorination in water treatment is to:

- A. Remove sediments
- B. Improve taste
- C. Kill pathogens
- D. Reduce turbidity

17. A freshwater food chain always begins with:

- A. Carnivores
- B. Decomposers
- C. Producers
- D. Omnivores

18. Which organism is a primary consumer in a freshwater ecosystem?

- A. Algae
- B. Zooplankton
- C. Fish
- D. Bacteria

19. A food web differs from a food chain because it:

- A. Shows only predators
- B. Is found only in lakes
- C. Shows multiple feeding relationships
- D. Includes only producers

20. Which interaction benefits one species while harming another?

- A. Mutualism
- B. Commensalism
- C. Competition
- D. Predation

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21. Competition in freshwater communities occurs when organisms:

- A. Eat different foods
- B. Share unlimited resources
- C. Require the same limited resources
- D. Live in different habitats

22. Which factor most directly limits population size in freshwater ecosystems?

- A. Population density
- B. Carrying capacity
- C. Birth rate
- D. Immigration

23. Carrying capacity refers to:

- A. Maximum growth rate
- B. Number of species present
- C. Maximum population an environment can support
- D. Total biomass

24. An invasive species is best described as one that:

- A. Is native and endangered
- B. Has no predators
- C. Is introduced and spreads rapidly
- D. Lives only in polluted water

25. Invasive species often cause problems because they:

- A. Increase biodiversity
- B. Have low reproduction rates
- C. Outcompete native species
- D. Improve water quality

26. A watershed is:

- A. A man-made water channel
- B. An underground aquifer
- C. An area of land that drains into a water body
- D. A water treatment facility

27. Activities in a watershed can affect water quality because:

- A. Water flows uphill
- B. Pollutants can be carried by runoff
- C. Only groundwater is affected
- D. Watersheds are isolated systems

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28. Which land use is MOST likely to increase sedimentation in streams?

- A. Forest preservation
- B. Wetland restoration
- C. Construction and deforestation
- D. Wildlife protection

29. Excess sediment in freshwater systems can:

- A. Increase light penetration
- B. Smother aquatic habitats
- C. Raise dissolved oxygen
- D. Reduce erosion

30. Turbidity refers to:

- A. Water temperature
- B. Water acidity
- C. Water clarity
- D. Nutrient concentration

31. High turbidity can negatively affect aquatic life by:

- A. Increasing photosynthesis
- B. Blocking sunlight
- C. Increasing oxygen levels
- D. Cooling the water

32. Which human activity MOST improves freshwater quality?

- A. Dumping waste
- B. Overfishing
- C. Riparian buffer restoration
- D. Channelization

33. Which organism is considered a top predator in many freshwater ecosystems?

- A. Algae
- B. Aquatic insects
- C. Large fish
- D. Zooplankton

34. Nitrogen enters freshwater ecosystems primarily through:

- A. Volcanic activity
- B. Fertilizer runoff and decomposition
- C. Evaporation
- D. Sedimentation

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35. Which process returns nutrients from dead organisms back into the ecosystem?

- A. Photosynthesis
- B. Respiration
- C. Decomposition
- D. Predation

36. Harmful algal blooms are often linked to:

- A. Low nutrient levels
- B. High water flow
- C. Excess nutrients
- D. Cold temperatures

37. Which practice helps protect watersheds?

- A. Removing vegetation
- B. Increasing impervious surfaces
- C. Reducing fertilizer use
- D. Straightening streams

38. Which freshwater organism is most sensitive to pollution?

- A. Trout
- B. Leeches
- C. Carp
- D. Mosquito larvae

39. Biological monitoring of freshwater often uses living organisms because they:

- A. Are easier to count
- B. Show long-term water quality conditions
- C. Replace chemical tests
- D. Live only in clean water

40. Which statement best describes freshwater sustainability?

- A. Using all available water
 - B. Protecting water resources for future generations
 - C. Eliminating aquatic species
 - D. Increasing pollution tolerance
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Part II: Freshwater Macroinvertebrates & Water Quality (Questions 41–50)

41. Macroinvertebrates are best described as:

- A. Microscopic fish
- B. Small vertebrates
- C. Aquatic animals without backbones visible to the naked eye
- D. Floating plants

42. Why are macroinvertebrates useful indicators of water quality?

- A. They are easy to remove
- B. They live only in lakes
- C. They differ in tolerance to pollution
- D. They reproduce quickly

43. Which macroinvertebrate indicates VERY clean, well-oxygenated water?

- A. Leech
- B. Midge larva
- C. Mayfly nymph
- D. Aquatic worm

44. Stonefly nymphs are typically found in:

- A. Warm, stagnant water
- B. Polluted ponds
- C. Fast-flowing, clean streams
- D. Urban drainage ditches

45. Which macroinvertebrate is moderately tolerant of pollution?

- A. Caddisfly larva
- B. Dragonfly nymph
- C. Stonefly nymph
- D. Mayfly nymph

46. Which organism is MOST tolerant of poor water quality?

- A. Mayfly
- B. Stonefly
- C. Leech
- D. Caddisfly

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47. Caddisfly larvae are unique because they often:

- A. Live on the water surface
- B. Build protective cases
- C. Breathe air
- D. Live only in lakes

48. A stream with many leeches and aquatic worms most likely has:

- A. Excellent water quality
- B. High dissolved oxygen
- C. Organic pollution
- D. Low nutrient levels

49. Which macroinvertebrate is a predator and moderately tolerant of pollution?

- A. Dragonfly nymph
- B. Mayfly nymph
- C. Stonefly nymph
- D. Water penny

50. A diverse macroinvertebrate community usually indicates:

- A. Recent flooding
 - B. Stable, healthy water conditions
 - C. High sediment levels
 - D. Severe pollution
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